CNC PROGRAM SHEET

Part: Part | Rev: A | Date: 2026-06-12

Programmer: CNC-AI

Material: ALUMINIUM

Max RPM: 4500 Coolant: through spindle

Setups: 3

Part envelope: 55.3 x 104.8 x 21.5 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face center of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	CD_60_2MM	60-degree combined center drill, 2mm tip, 5mm body	2.0	Sandvik Coromant	H10F	spot_drill
T02	EM_2FL_2MM	2-flute solid carbide end mill 2mm â€” aluminium geometry	2.0	Sandvik Coromant	1630	contour_mill
T03	DRILL_2.5MM	Solid carbide jobber drill 2.5mm	2.5	Sandvik Coromant	H10F	twist_drill
T04	DRILL_3.0MM	Solid carbide jobber drill 3.0mm	3.0	Sandvik Coromant	H10F	twist_drill
T05	DRILL_3.2MM	Solid carbide jobber drill 3.2mm â€” standard size for Ã˜3.2mm pin holes	3.2	Sandvik Coromant	H10F	twist_drill
T06	DRILL_4.0MM	Solid carbide jobber drill 4.0mm	4.0	Sandvik Coromant	H10F	twist_drill
T07	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T08	BALL_2FL_AL_4MM	2-flute solid carbide ball-nose end mill 4mm â€” aluminium geometry	4.0	Sandvik Coromant	1630	profile_3d_contour
T09	DRILL_5.0MM	Solid carbide jobber drill 5.0mm	5.0	Sandvik Coromant	H10F	twist_drill
T10	SPOT_090_6MM	90-degree solid carbide spot drill, 6mm shank	6.0	Sandvik Coromant	H10F	spot_drill
T11	EM_2FL_6MM	2-flute solid carbide end mill 6mm â€” aluminium geometry	6.0	Sandvik Coromant	1630	pocket_mill
T12	CHAM_90_6MM	90-degree solid carbide chamfer mill 6mm, 2-flute	6.0	Sandvik Coromant	1630	chamfer_mill
T13	DRILL_10.0MM	Solid carbide jobber drill 10.0mm â€” also used as pilot drill	10.0	Sandvik Coromant	H10F	pilot_drill, twist_drill
T14	DRILL_16.0MM	Solid carbide jobber drill 16.0mm â€” also used as core drill	16.0	Sandvik Coromant	H10F	core_drill
T15	EM_2FL_20MM	2-flute solid carbide end mill 20mm â€” aluminium geometry	20.0	Sandvik Coromant	1630	circular_interp
T16	FACEMILL_40MM	40mm indexable face mill â€” 6 inserts	40.0	Sandvik Coromant		face_mill
T17	FACEMILL_50MM	50mm indexable face mill / shell mill â€” 4 inserts	50.0	Sandvik Coromant		face_mill

SETUP 1 -- -Z approach (rear face)

PRINCIPAL | 42 clusters | 72 operations

Fixture	90° angle plate or tombstone fixture.
Description	Rear-face setup. Rotate part 90° so rear face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=0.000mm) Y=CENTER (Y=-20.711mm) Z=TOP
Face (W x H)	55.3 x 104.8 mm
Max depth	21.5 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
2	ALU 4 DRILL	T06	4.0	0	0.41	1590	95	0.408	--	--	00:01
3	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
4	ALU 4 DRILL	T06	4.0	0	0.41	1590	95	0.408	--	--	00:01
5	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
6	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
7	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
8	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
9	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
10	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
11	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
12	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
13	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
14	ALU 4 DRILL	T06	4.0	0	0.33	1590	95	0.332	--	--	00:01
15	ALU 50 FACEMILL BORE FINISH	T17	50.0	0	0.29	4500	2700	0.294	0.0	0.0	00:01
16	ALU 20 ENDMILL BORE	T15	20.0	0	0.29	4500	540	0.294	--	--	00:10
17	ALU 50 FACEMILL BORE FINISH	T17	50.0	0	0.29	4500	2700	0.294	0.0	0.0	00:01
18	ALU 20 ENDMILL BORE	T15	20.0	0	0.29	4500	540	0.294	--	--	00:10
19	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
20	ALU 4 DRILL	T06	4.0	0	0.33	1590	95	0.332	--	--	00:01
21	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
22	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
23	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
24	ALU 4 DRILL	T06	4.0	0	0.38	1590	95	0.375	--	--	00:01
25	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
26	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
27	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
29	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
30	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
31	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
32	ALU 4 DRILL	T06	4.0	0	0.02	1590	95	0.019	--	--	00:01
33	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
34	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
35	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
36	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
37	ALU CENTER DRILL	T10	6.0	0	3.0	420	21	3.0	--	--	00:09
38	ALU 10 DRILL	T13	10.0	0	1.5	4500	495	1.5	--	--	00:01
39	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
40	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
41	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
42	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
43	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
44	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
45	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
46	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
47	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
48	ALU 3.2 DRILL	T05	3.2	0	1.5	1190	80	1.5	--	--	00:01
49	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
50	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
51	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
52	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
53	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
54	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
55	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
56	ALU 2.5 DRILL	T03	2.5	0	2.5	1760	83	2.5	--	--	00:02
57	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.01	4500	90	0.009	0.1	0.1	00:06
58	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.01	4500	90	0.009	0.0	0.0	00:06
59	ALU 2 ENDMILL BOSS RF	T02	2.0	0	0.02	4500	90	0.019	0.1	0.1	00:06
60	ALU 2 ENDMILL BOSS FINISH	T02	2.0	0	0.02	4500	90	0.019	0.0	0.0	00:06
61	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.03	3580	358	0.025	0.0	0.0	00:10
62	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
63	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
64	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:10
65	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.81	3580	358	0.4	0.0	0.0	00:10
66	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
67	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.38	3580	358	0.375	0.0	0.0	00:10
68	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:10
69	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.62	3580	358	0.4	0.0	0.0	00:10
70	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:10
71	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.88	3580	358	0.4	0.0	0.0	00:10
72	ALU 40 FACEMILL FACE FINISH	T16	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05

ESTIMATED SETUP TIME: 05:45 (includes tool changes)

SETUP 2 -- +Y approach (top — default VMC position)

PRINCIPAL | 16 clusters | 23 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (X=0.000mm, 27.67mm from xmin edge) Y=CENTER (Z=2.750mm, 10.76mm from zmin edge) Z=TOP
Face (W x H)	55.3 x 21.5 mm
Max depth	104.8 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
2	ALU 5 DRILL	T09	5.0	0	9.99	1400	98	9.988	--	--	00:06
3	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
4	ALU 5 DRILL	T09	5.0	0	9.4	1400	98	9.405	--	--	00:06
5	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
6	ALU 5 DRILL	T09	5.0	0	9.4	1400	98	9.405	--	--	00:06
7	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
8	ALU 5 DRILL	T09	5.0	0	9.99	1400	98	9.988	--	--	00:06
9	ALU CENTER DRILL	T10	6.0	0	3.0	420	21	3.0	--	--	00:09
10	ALU 10 DRILL	T13	10.0	0	17.03	4500	540	17.033	--	--	00:02
11	ALU 16 DRILL	T14	16.0	0	17.03	3580	573	17.033	--	--	00:02
12	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	2.54	3580	358	0.4	0.0	0.0	00:10
13	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.05	3580	358	0.051	0.0	0.0	00:10
14	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	4.1	3580	358	0.4	0.0	0.0	00:10
15	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	2.37	3580	358	0.4	0.0	0.0	00:10
16	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	4.27	3580	358	0.4	0.0	0.0	00:10
17	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.05	3580	358	0.051	0.0	0.0	00:10
18	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
19	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
20	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
21	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	9.62	3580	358	0.4	0.0	0.0	00:10
22	ALU 6 ENDMILL POCKET RF	T11	6.0	0	--	3020	483	6.0	0.1	0.1	00:05
23	ALU 6 ENDMILL POCKET FINISH	T11	6.0	0	--	3020	483	6.0	0.0	0.0	00:05

ESTIMATED SETUP TIME: 03:44 (includes tool changes)

SETUP 3 -- -X approach (left side face)

PRINCIPAL | 12 clusters | 14 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (Y=-20.711mm) Y=CENTER (Z=2.750mm) Z=TOP
Face (W x H)	104.8 x 21.5 mm
Max depth	55.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T07	4.0	0	2.0	640	32	2.0	--	--	00:04
2	ALU 5 DRILL	T09	5.0	0	0.74	1400	98	0.738	--	--	00:01
3	ALU CENTER DRILL	T01	2.0	0	1.0	950	76	1.0	--	--	00:01
4	ALU 3 DRILL	T04	3.0	0	3.57	1170	78	3.573	--	--	00:03
5	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
6	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
7	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
8	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	0.49	3580	358	0.4	0.0	0.0	00:10
9	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	3.0	3580	358	0.4	0.0	0.0	00:10
10	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	4.75	3580	358	0.4	0.0	0.0	00:10
11	ALU 4 BALLNOSE FILLET FINISH	T08	4.0	0	1.69	3580	358	0.4	0.0	0.0	00:10
12	ALU 40 FACEMILL FACE FINISH	T16	40.0	0	2.0	4500	3240	0.5	0.0	0.0	00:05
13	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	4500	540	2.0	--	--	00:10
14	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	4500	540	2.0	--	--	00:10

ESTIMATED SETUP TIME: 02:40 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
SUBSTITUTION	C2	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C3	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C7	face_mill	Effective Vc = 706.9 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	circular_interp	Effective Vc = 282.7 m/min (reduced from 500 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	face_mill	Effective Vc = 706.9 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	circular_interp	Effective Vc = 282.7 m/min (reduced from 500 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C18	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C19	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C20	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
SUBSTITUTION	C21	twist_drill	SUBSTITUTION: 4.6mm is non-standard. Using nearest standard size 4.5mm. Verify hole tolerance before confirming. Conser
WARNING	C22	spot_drill	WARNING: no spot drill covers dia=16.1mm, using largest available 6.0mm Field-validated: RPM ~425 F21 â€” slow spot for p
SUBSTITUTION	C22	pilot_drill	SUBSTITUTION: 9.66mm is non-standard. Using nearest standard size 10.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C22	pilot_drill	Effective Vc = 141.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C22	core_drill	SUBSTITUTION: 16.1mm is non-standard. Using nearest standard size 16.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C25	twist_drill	Effective Vc = 141.4 m/min (reduced from 220 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C31	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C32	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C33	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
SUBSTITUTION	C34	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming. Field
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 28.3 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C65	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C67	face_mill	Effective Vc = 565.5 m/min (reduced from 800 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C68	chamfer_mill	Effective Vc = 84.8 m/min (reduced from 85 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C69	chamfer_mill	Effective Vc = 84.8 m/min (reduced from 85 m/min). Feed rate adjusted accordingly.